INCH-POUND

MIL-A-63991 (AR)
AMENDMENT 1
28 July 1995

MILITARY SPECIFICATION

ALTERNATOR, TURBINE (AIR DRIVEN)

This amendment forms a part of MIL-A-63991 (AR), dated 28 March 1988, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2, Under PRODUCT AND PACKAGING DRAWINGS, add the following:

"11744334 - Alternator, Turbine (Air Driven)".

PAGE 3

3.3, Line 3: Delete "11722760" and substitute "11722760/11744334".

PAGE 4

3.3.3, Line 2: Delete "11722760" and substitute "11722760/11744334".

3.6.1, Line 1: Delete "11722761" and substitute "11722761/11744331".

AMSC N/A FSC 1390

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PAGE 6

4.3.1, Under PRODUCT AND PACKAGING DRAWINGS: Delete old drawing numbers and substitute new drawing numbers as follows:

"OLD DWG. NO.	NEW DWG. NO.
11722760	11722760/11744334
11722766	11722766/11744341
11722765	11722765/11744340
11722764	11722764/11744338
11722763	11722763/11744339
11730640	11730640/11744331
11730641	11730641/11744327
11722761	11722761/11744337
11730642	11730642/11744328
11730643	11730643/11744329
11723110	11723110/11744332
11723111	11723111/11744333
11730638	11730638/11744330
11722762	11722762/11744342".

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Table I, Under PRODUCT AND PACKAGING DRAWINGS: Delete old drawing numbers and substitute new drawing numbers as follows:

"OLD DWG. NO.	NEW DWG. NO.
11722760	11722760/11744334
11722766	11722766/11744341
11722765	11722765/11744340
11722764	11722764/11744338
11722763	11722763/11744339".

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Table I, Under PRODUCT AND PACKAGING DRAWINGS: Delete old drawing numbers and substitute new drawing numbers as follows:

"OLD DWG. NO.	NEW DWG. NO.
11730640	11730640/11744331
11730641	11730641/11744327
11722761	11722761/11744337
11730642	11730642/11744328
11730643	11730643/11744329
11723110	11723110/11744332".

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Table I, Under PRODUCT AND PACKAGING DRAWINGS: Delete old drawing numbers and substitute new drawing numbers as follows:

"OLD DWG. NO.	NEW DWG. NO.
11723111	11723111/11744333
11730638	11730638/11744330
11722762	11722762/11744342".

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4.4.2.1: Delete "11722760" and substitute "11722760/11744334".

PAGE 13

4.4.2.2: Delete "11722766" and substitute "11722766/11744341".
Delete "11722760" and substitute "11722760/11744334".

PAGE 14

4.4.2.3: Delete "11722765" and substitute "11722765/11744340".
Delete "11722766" and substitute "11722766/11744341".

PAGE 15

4.4.2.4: Delete "11722764" and substitute "11722764/11744338".
Delete "11722765" and substitute "11722765/11744340".

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4.4.2.5: Delete "11722763" and substitute "11722763/11744339".
Delete "11722765" and substitute "11722765/11744340".

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4.4.2.6.1: Add new CD sheet, page 17a, as follows:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6.1	Shaft and Magnet Assembly			11744331
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 11744334 / Inspection Method Reference
CRITICAL	None defined			
MAJOR				
101	Runout between magnet and shaft (2 axis)	0.40%	3.3	Gage
102	Runout of magnet faces to shaft	0.40%	3.3	Gage
103	Distance from aft face of washer to aft face of magnet	0.40%	3.3	Gage
MINOR				
201	Distance from aft shoulder of bearing surface to aft face of magnet	0.65%	3.3	Gage
202	Evidence of surface defects on raceways and datum A surfaces	0.65%	3.3	Gage
203	Magnet and shaft pull test	0.65%	3.6	4.5.7/ Visual
204	Evidence of poor workmanship	0.65%	3.9	Visual

NOTES:

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4.4.2.7.1: Add new CD sheet, page 18a, as follows:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7.1	Shaft			11744327
				NEXT HIGHER ASSEMBLY 11744331
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	Inspection Method Reference
CRITICAL	None defined			
MAJOR				
101	Surface finish of bearing raceway	0.40%	3.3	Gage
102	Length of aft shaft bearing surface (.085)	0.40%	3.3	Gage
103	Distance from aft face of washer to aft face of magnet	0.40%	3.3	Gage
104	Bearing surface diameter (2 places)	0.40%	3.3	Gage
105	Improper angle cut of aft "V" slot	0.40%	3.3	Gage
MINOR				
201	Improper width of aft "V" slot	0.65%	3.3	Gage
204	Evidence of poor workmanship	0.65%	3.9	Visual
NOTES:				

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4.4.2.8: Delete "11722761" and substitute "11722761/11744337". Delete "11730640" and substitute "11730640/11744331".

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4.4.2.9.1: Add new CD sheet, page 20a, as follows:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9.1	Housing			11744328
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 11744332 / Inspection Method Reference
CRITICAL	None defined			
MAJOR				
101	Diameter of bearing cavity (.1866)	0.40%	3.3	Gage
102	Height from inner housing base to shelf	0.40%	3.3	Gage
103	Inner housing shelf diameter (1.128)	0.40%	3.3	Gage
MINOR				
201	Inside flange leg diameter	0.65%	3.3	Gage
202	Inside diameter of bearing shaft cavity (.1866)	0.65%	3.3	Gage
203	MMC of cutouts incorrect	0.65%	3.3	Gage
204	Evidence of poor workmanship	0.65%	3.9	Visual

NOTES:
1. The same 50 samples shall be used for all examinations.
2. A new sample shall be drawn from each lot, sampling run or fixture.
3. If two or more defects are found, the housing lot shall be rejected or screened in its entirety of the defect(s) and be resubmitted for inspection.

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4.4.2.10.1: Add new CD sheet, page 21a, as follows:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10.1	End Plate			11744329
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 11744333 / Inspection Method Reference
CRITICAL	None defined			
MAJOR				
101	Inside diameter of bearing cavity	50	3.3	Gage
102	Outside diameter of end plate	50	3.3	Gage
MINOR				
201	Inside flange leg diameter	50	3.3	Gage
202	MMC of cutouts incorrect	50	3.3	Gage
203	Evidence of poor workmanship	50	3.9	Visual

NOTES:
1. The same 50 samples shall be used for all examinations.
2. A new sample shall be drawn from each lot, sampling run or fixture.
3. If two or more defects are found, the housing lot shall be rejected or screened in its entirety of the defect(s) and be resubmitted for inspection.

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4.4.2.11: Add new Major 104 as follows:

"104 .045 height ... 0.40% ... 3.3 ... Gage".

4.4.2.11.1: Add new CD sheet, page 22a, as follows:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER
4.4.2.11.1	Housing and Bearing Assembly	SHEET 1 OF 1		11744332
				NEXT HIGHER ASSEMBLY 11744334
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	Inspection Method Reference
CRITICAL	None defined			
MAJOR				
101	Exceeds the .018 maximum dimension	0.40%	3.3	Gage
102	Bearing not fully seated	0.40%	3.3	Gage
MINOR				
201	Evidence of poor workmanship	0.65%	3.9	Visual
NOTES:				

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4.4.2.12.1: Add new CD sheet, page 23a, as follows:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH				DRAWING NUMBER
4.4.2.12.1	End Plate and Bearing Assembly	SHEET 1 OF 1		11744333
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 11744334 / Inspection Method Reference
CRITICAL	None defined			
MAJOR				
101	Exceeds the .018 maximum dimension	0.40%	3.3	Gage
102	Bearing not fully seated	0.40%	3.3	Gage
MINOR				
201	Evidence of poor workmanship	0.65%	3.9	Visual
NOTES:				

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4.4.2.13: Delete "11723115" and substitute "11723115/11744344". Delete "11730638" and substitute "11730638/11744330".

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4.4.2.14.1: Add new CD sheet, page 25a, as follows:

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			DRAWING NUMBER
4.4.2.14.1	Bearing-Flange Open Ball	SHEET 1 OF 1		11744330
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY 11744332/ 11744333 Inspection Method Reference
CRITICAL	None defined			
MAJOR				
101	Exceeds the .018 maximum dimension	0.40%	3.3	Gage
102	Bearing not fully seated	0.40%	3.3	Gage
MINOR				
201	Evidence of poor workmanship	0.65%	3.9	Visual
NOTES:				

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4.4.2.15: Delete "11722762" and substitute "11722762/11744342". Delete "11722760" and substitute "11722760/11744334".

PAGE 27

4.5.3: Delete "11730640" and substitute "11730640/11744341".

4.5.4: Delete "11722761" and substitute "11722761/11744337".

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4.5.7, Line 8: Delete "11722762" and substitute "11722762/11744342".

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PAGE 30

6.7: Delete "Alinbal, Inc.
28 Woodmont Rd.
Milford, CT 06460" and substitute

"Alinbal, Inc.
28 Woodmont Road
Milford, CT 06460
FSCM 31633

EECO, Inc.
1601 Chestnut Avenue
Santa Ana, CA 92702
FSCM 97525

Eastman Kodak Company
121 Lincoln Avenue
Rochester, NY 14611
FSCM 32153".

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6.7.1, Line 3: Delete "11722760" and substitute "11722760/11744334".

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1390-A686)

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